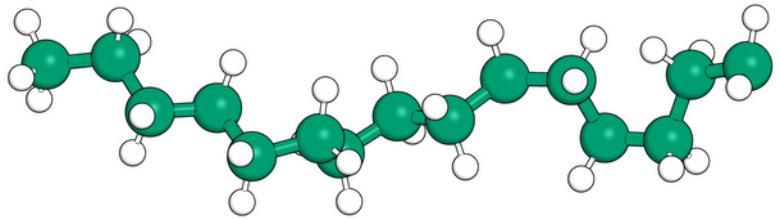
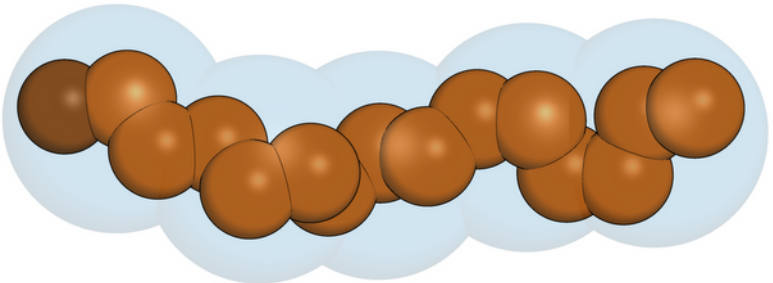
(a) AA (OPLS): C + explicit H



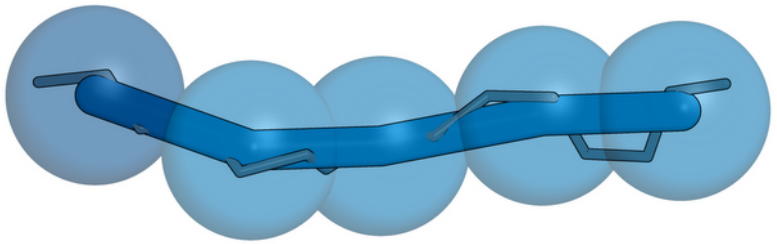
15 backbone C, 31 H

(b) UA (TraPPE): CH_2/CH_3 sites



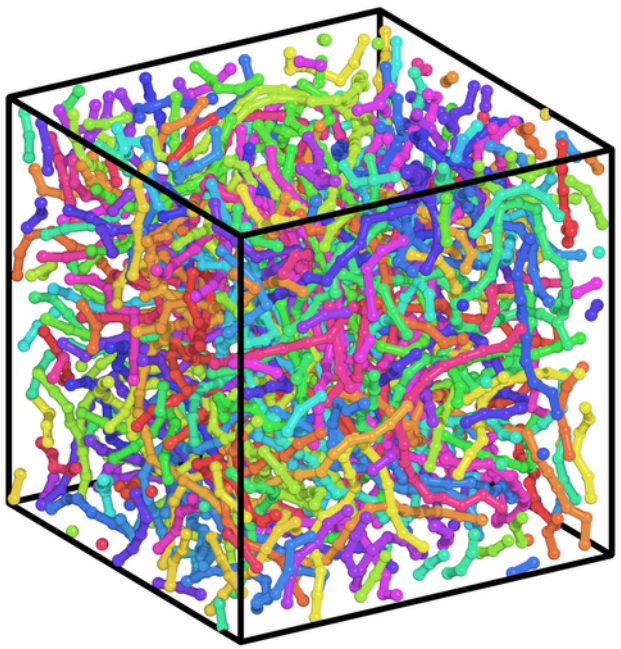
15 sites; faint blue = CG beads

(c) CG (SDK): 3-carbon beads



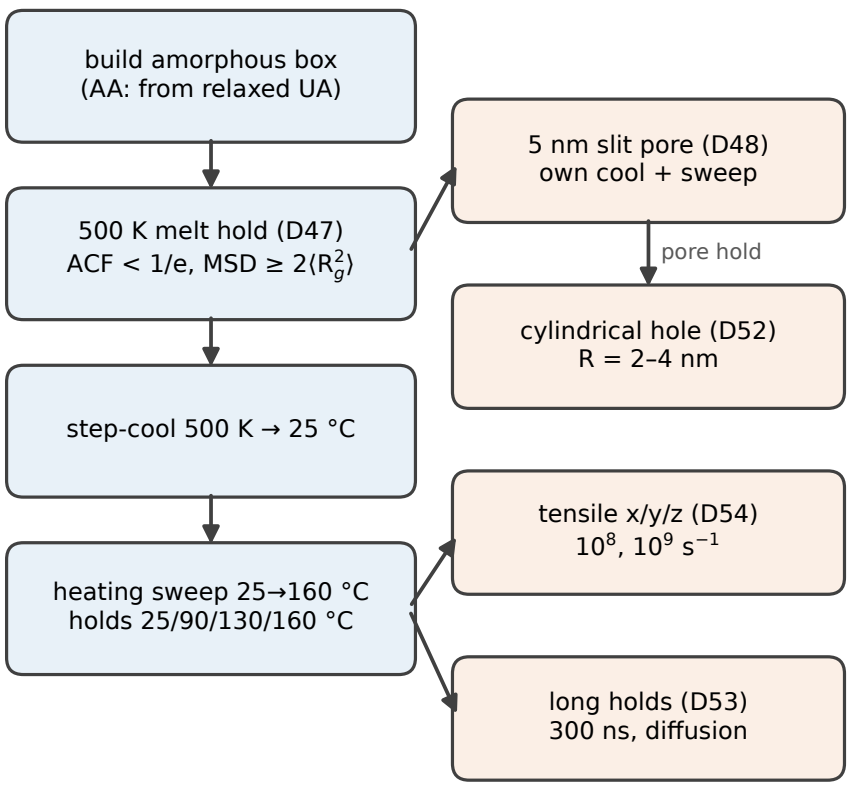
5 beads of the same segment

(d) CG PE bulk box



40 chains, 3,200 beads, 6.4 nm; 25 °C hold end

(e) Workflow



PE and iPP at CG (PE only), UA, AA; branches start from snapshots

(f) Systems: size and cost

system	sites	box (nm)	ns	ns/day
CG PE 40×C240	3,200	6.4³	1,109	233
CG PE 40×C240, 5 nm pore	3,200	6.7²×11.5	212	252
CG PE 320×C240	25,600	13.5³	73	280
CG PE 320×C240, 5 nm pore	25,600	13.5²×17.9	212	287
CG PE 640×C240	51,200	27.0²×6.7	0.24	19
CG PE 40×C1500	20,000	11.9³	772	312
UA PE 40×C240	9,600	6.7³	156	13
AA PE 40×C240	28,880	6.7³	7	5
UA iPP 40×C240	12,760	6.5³	121	14
AA iPP 40×C240	28,880	6.7³	7	5